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# Quenching timescales of dwarf satellites around Milky Way-mass hosts

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## ABSTRACT

We present measurements of the quenched fractions and quenching timescales of dwarf satellite galaxies, derived from a suite of 4 ultra high-resolution cosmological simulations of Milky Way-mass halos and a complementary sample of field dwarf galaxies. We show that these simulations accurately reproduce the satellite luminosity functions of observed nearby galaxies, as well as variations in quenched fractions across distance and stellar mass scales. In particular, we find that few dwarfs with $10^7 < M_*/\mathrm{M}_\odot < 10^9$ are quenched in the field, whereas $\sim 50\%$ of satellites in this stellar mass range are quenched. We then trace the histories of satellite galaxies back to $z \sim 15$, finding that the majority of galaxies in our sample with $M_* < 10^5$ $\mathrm{M}_\odot$ are consistent with quenching due to reionization $\gtrsim 11$ Gyr ago. Many galaxies which survive reionization ($M_*/\mathrm{M}_\odot > 10^6$) quench rapidly following infall, with quenching timescales of 1–2 Gyr, indicating that ram pressure stripping is likely the dominant environmental quenching mechanism. Finally, we find a sharp stellar mass threshold of $10^8$ $\mathrm{M}_\odot$, above which infalling satellites are largely resistant to rapid environmental quenching processes.

*Keywords:* galaxies: evolution – galaxies: star formation – galaxies: dwarf – galaxies: luminosity function – cosmology: reionization

## 1. INTRODUCTION

The gaseous halo environment of massive central galaxies has been the subject of extensive study and has been shown to have significant impacts on the formation and evolution of nearby dwarf satellites. In particular, observations of dwarf galaxies in the Local Group (LG) have shown trends in star-formation and gas properties across wide environmental and luminosity ranges. Surveys of the nearby universe have shown that isolated dwarfs with $10^7$ $\mathrm{M}_\odot < M_* < 10^9$ $\mathrm{M}_\odot$ are predominantly star-forming (Geha et al. 2012). In contrast, nearly all LG dwarf satellites with $M_* < 10^8$ $\mathrm{M}_\odot$ are quenched, or no longer actively forming stars (Wetzel et al. 2015b; Phillips et al. 2014; Weisz et al. 2015). This dichotomy is supported further by observations of the gas content of LG dwarfs, with atomic hydrogen (H I) undetected in all satellites within $\sim$ 270 kpc of their host but prevalent at higher distances (Grcevich & Putman 2009; Spekkens et al. 2014). Thus, it is clear that the quenching of dwarf galaxies is highly dependent upon environment, but it is unclear the exact processes which produce this dichotomy.

Traditionally, gas-loss due to ram pressure and tidal stripping has been the predominant explanation of environmental quenching, in which accretion onto the host halo removes star-forming gas from the disk of the satellite (Mayer et al. 2006). Ram pressure stripping has been shown to efficiently quench dwarfs in many simulations (Slater & Bell 2014; Wheeler et al. 2014; Simpson et al. 2018); however, Emerick et al. (2016) used high-resolution wind tunnel simulations to show that ram pressure may be unable to entirely quench galaxies as quickly as was believed. Additionally, ram pressure has been shown to be highly efficient at removing hot gas, particularly outside of the satellite's disk, but to be less efficient at removing cold, clumpy gas (Bahé et al. 2013; Hausammann et al. 2019). Hausammann et al. (2019) argue that simulations that prevent gas from cooling below the critical threshold of $\sim 10^4$ K are prone to overestimating the quenching efficiency of ram pressure. Thus, ram pressure and tidal stripping alone are likely not the only drivers of environmental quenching.

Stellar feedback processes such as supernovae winds have been shown to play a significant role in the evolution of low-mass satellites, both by increasing the effi-

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ciency of stripping by expelling and heating central gas (Bahé & McCarthy 2015; Kazantzidis et al. 2017) and through reducing the central satellite mass and therefore reducing its potential (Zolotov et al. 2012). Additionally, at higher masses, strangulation via suppressed accretion, gas depletion, and feedback-driven outflows may be able to efficiently quench satellites (McGee et al. 2014). Gas depletion, SN feedback, and ram pressure stripping may all conspire to quench satellites, with ram pressure removing SN-heated gas and preventing gas recycling (Bahé et al. 2013). Finally, at the very lowest mass scales ($M_* < 10^5$ M$_\odot$), star formation is typically suppressed very early in cosmic time, consistent with quenching due to photoevaporation of gas during the reionization of the early universe (Bullock et al. 2000; Brown et al. 2014). These processes appear to be highly interconnected, rendering studies of each in isolation inadequate to determine their relative roles.

In order to disentangle the roles of these various quenching processes, many studies have estimated the quenching timescales necessary to reproduce the high observed quenched fractions in the Local Group, predominantly finding that quenching must occur rapidly (1–2 Gyr) after infall for low-mass ($M_* < 10^8$ M$_\odot$) satellites (Slater & Bell 2014; Fillingham et al. 2015; Wetzel et al. 2015b). While these results hint at a highly efficient process, if ram pressure alone cannot quench satellites on this timescale (Emerick et al. 2016), there must be additional processes at play. For more massive satellites, in which ram pressure stripping is known to be largely inefficient, estimates show much longer quenching timescales ($\gtrsim$ 8 Gyr), more consistent with strangulation models in which quenching occurs on the cold gas depletion timescale (Wheeler et al. 2014; Fillingham et al. 2015; Balogh et al. 2016).

The vast majority of these works use $N$-body smooth particle hydrodynamic or adaptive mesh refinement simulations to follow gas flow, constrain infall times, and make claims about the relative importance of quenching processes. However, many of these simulations achieve high spatial and mass resolution only by focusing on individual satellites (e.g. Mayer et al. 2006; Emerick et al. 2016). Others that simulate larger cosmological volumes do so at the cost of resolution (e.g. Bahé & McCarthy 2015). Additionally, many of these studies use abundance-matching techniques and dark matter-only simulations to constrain infall times of Local Group satellites (Wetzel et al. 2015b; Fillingham et al. 2015, e.g.). Dark matter-only simulations have been susceptible to such problems as the Too Big to Fail, Core-Cusp, and Missing Satellite problems (Boylan-Kolchin et al. 2011; Moore et al. 1999), but the inclusion of stellar feedback in simulations has been shown to help resolve these problems (Brooks et al. 2013). Only recently have high resolution simulations containing realistic star formation, stellar feedback, and gas physics been introduced and implemented in studies of dwarf galaxies, largely producing satellite properties consistent with observations (e.g. Simpson et al. 2018; Garrison-Kimmel et al. 2019a,b).

In this paper, we use the recently completed DC Justice League suite of ultra-high resolution cosmological simulations to analyze the relative importance of quenching mechanisms in MW-mass halo environments. The structure of the paper is as follows. In §2, we present the suite of simulations. In §3 we explore the luminosity distribution and star formation histories of resolved halos at $z = 0$. In §4 we turn our attention beyond $z = 0$, tracing halos back in time and examining their infall, quenching, and gas-loss times. In §5 we discuss the relative impact of various quenching processes, and we present our conclusions in §6.

## 2. METHODS

We conduct our analysis primarily on the DC Justice League suite of 4 high-resolution, cosmological, smooth particle hydrodynamics (SPH) simulations of MW-analog regions, each containing a massive ($M_{\rm vir} \sim 10^{12}$ M$_\odot$) host halo. We complement these simulations with the Marvel-ous suite of 4 low-density regions, yielding a sample of 174 resolved, luminous dwarf galaxies for analysis. Here we discuss the simulation code and the post-processing analysis conducted on the simulation output. Basic properties of the host halos in the Justice League simulations are shown in Table 1.

### 2.1. *Simulations*

All simulations are generated from the tree+SPH code ChaNGa (Menon et al. 2015), which scales efficiently up to 100,000 cores. ChaNGa is the successor to the N-body tree+SPH code gasoline. Simulations are integrated to redshift 0 in a fully cosmological context assuming Plank 2015 cosmological parameters ($\Omega_0 = 0.3086$, $\Lambda = 0.6914$, $h = 0.67$, $\sigma_8 = 0.77$; Planck Collaboration 2016). In order to achieve ultra-high resolution while maintaining the effects of the large-scale environment, we use "zoom-in" volume renormalization tehcnique (Katz & White 1993) with initial conditions generated from $50^3$ Mpc$^3$ dark matter-only volumes. The simulations have a force resolution of 87 pc and gas/star particle masses of 3300/994 M$_\odot$, allowing us to resolve halos down to $M_{\rm vir} \sim 10^6$ M$_\odot$.

ChaNGa follows non-equilibrium abundances of H and He species. Photoionization and heating rates are

implemented using a uniform, time-dependent cosmic UV background adapted from Haardt & Madau (2012). In this model, cosmological $H_2$ regions overlap at $z \sim 6.7$ (13 Gyr ago), indicating the onset of reionization. Star-formation in ChaNGa is implemented probabilistically according to local $H_2$ abundance, gas density, and gas temperature. This star-formation model is described in detail in Christensen et al. (2012), but a short description is given here. Gas particles that are sufficiently dense ($\rho \geq 0.1$ a.m.u. cm$^{-3}$) and cool ($T \leq 1000$ K) have a given probability $p$ of spawning a star particle. The probability $p$ is a function of the local dynamical time $t_{\rm form}$:

$$p = \frac{m_{\rm gas}}{m_{\rm star}} \left(1 - e^{-c^* \Delta t / t_{\rm form}}\right), \quad (1)$$

where $m_{\rm gas}$ is the mass of the gas particle, $m_{\rm star}$ is the initial mass of the potential star particle, and $c^*$ is the star-forming efficiency factor and is related to the $H_2$ fraction by $c^* = 0.1 X_{\rm H_2}$. This model also includes dust and $H_2$ self-shielding, as in Christensen et al. (2012).

Energy from Type II supernovae (SNe II) is distributed as thermal energy to surrounding gas particles according to the "blastwave" model (Stinson et al. 2006), assuming the canonical $\eta_{\rm SN} = 1.5 \times 10^{51}$ ergs released per SN. Cooling of the affected gas particles is temporarily disabled to match the theoretical timescale of the "snowplow" phase (McKee & Ostriker 1977) and to avoid rapid radiative cooling that would render feedback negligible. This feedback model has been proven realistic, reproducing observed properties such as cored dark matter profiles in dwarf galaxies (Governato et al. 2010), bar formation in massive spirals (Zana et al. 2019), the mass–metallicity relation and the scaling of the mass loading factor (Christensen et al. 2016), and a solution to the too big to fail problem (Brooks & Zolotov 2014).

Additional physics modules in ChaNGa include cooling from metal lines and metal diffusion (Shen et al. 2010) and supermassive black hole formation, growth, and feedback (Tremmel et al. 2017).

### 2.2. *Post-processing Analysis*

In order to select individual halos from simulation snapshots, we used amiga's halo finder (Knollmann & Knebe 2009), which identifies regions of overdensity and assigns halo ownership to gravitationally bound particles. We compute the virial radius ($R_{\rm vir}$) of each halo as the radius at which the enclosed density drops below 200 times the background density, $\rho_b$. Halo properties are calculated from all particles within the virial radius, excluding subhalos. Table 1 shows basic $z = 0$ properties of the host halos in the Justice League simulations, and Figure 1 shows line-of-sight averaged density plots of the dark matter, gas, and stars in Sandra.

**Table 1.** Host Halo Properties

| Sim | $M_{\rm vir}$ (M$_\odot$) | $M_*$ (M$_\odot$) | $M_{\rm gas}$ (M$_\odot$) | $R_{\rm vir}$ (kpc) |
|---|---|---|---|---|
| Sandra | $2.7 \times 10^{12}$ | $1.9 \times 10^{11}$ | $2.0 \times 10^{11}$ | 432 |
| Ruth | $1.3 \times 10^{12}$ | $1.0 \times 10^{11}$ | $9.3 \times 10^{10}$ | 340 |
| Sonia | $1.2 \times 10^{12}$ | $9.0 \times 10^{10}$ | $9.7 \times 10^{10}$ | 333 |
| Elena | $8.3 \times 10^{11}$ | $9.0 \times 10^{10}$ | $3.5 \times 10^{10}$ | 293 |

For each Justice League simulation, we used the database-generation software tangos (Pontzen & Tremmel 2018) to track particles across snapshots and generate merger trees for all halos containing stars at $z = 0$. For each halo, we determine the major progenitors, defined as the halo which contains the majority of particles from the corresponding halo in the subsequent snapshot, back to $z \approx 15$. The time resolution of all resulting data is constrained by the mean difference between timesteps, $\sim 300$ Myr. Additionally, by default tangos only tracks halos with $> 1000$ dark matter particles, corresponding to a halo mass of $\sim 10^6$ M$_\odot$, approximately the resolution limit of our simulations. All further analysis was conducted in Python using the pynbody package (Pontzen et al. 2013).

## 3. HALO PROPERTIES

### 3.1. *Luminosity Distribution*

The left panel of Figure 2 shows the cumulative satellite luminosity functions (LFs) for our four Justice League simulations, alongside observed LFs for nearby host-satellite systems. The LFs of our simulations vary widely, from $\sim 3$ to $\sim 15$ satellites at $M_V = -10$, but are generally ordered by host halo mass (see Table 1). That is, on the faint end ($M_V > -11$), the more massive halo hosts more luminous satellites. This trend is to be expected, as the number of dwarf satellites is known to be correlated to the mass of the host halo (Trentham & Tully 2009; Nickerson et al. 2013). However, this ordering by host mass does not remain true for bright satellites. The LF for Ruth drops off significantly at $M_V < -14$, despite the host being roughly equal in mass to Sonia and far more massive than Elena. This drop-off is also seen in the mass function (i.e. there are no dark-matter halos with $M_{\rm vir} > 3 \times 10^9$ M$_\odot$ in Ruth), indicating that this is not an issue with star-formation.

Despite the widely varying LFs of our simulations, they are consistent with recently observed LFs in nearby galaxies. Our most populous system, Sandra, is consistent with the LFs of massive galaxies such as Centaurus A, M31, and M81. Our less massive hosts are most con-

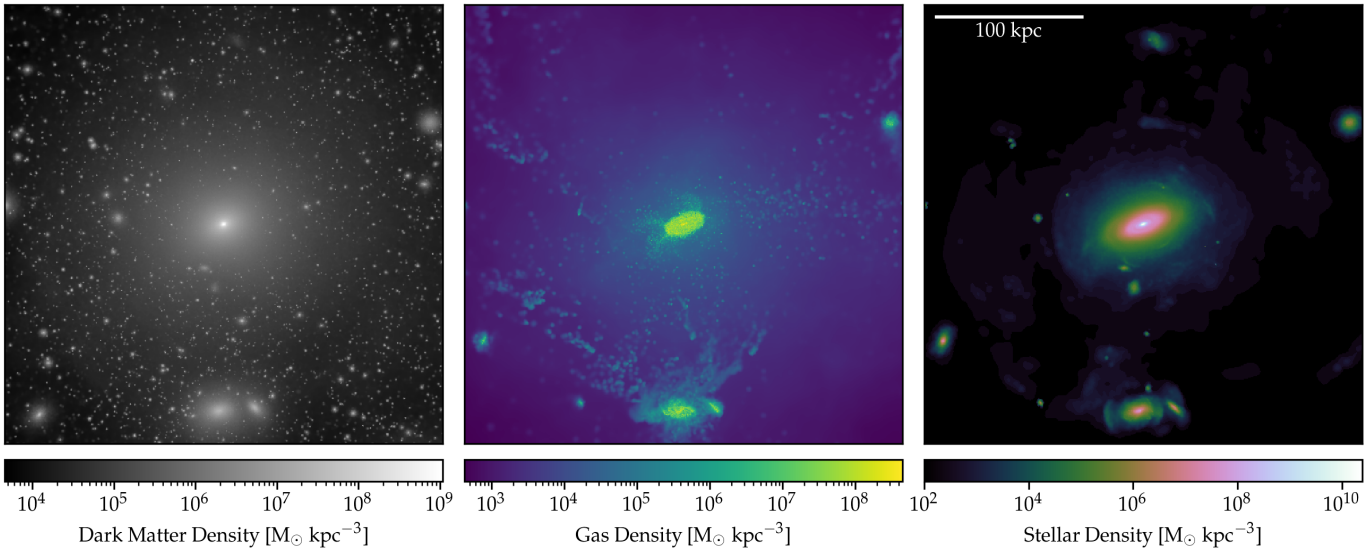


**Figure 1.** Line-of-sight averaged density plots of Sandra. Several dwarf satellites can be seen, many of which are gas-poor.

sistent with the MW and M101, and our least massive host, Elena, is highly consistent with the unique and sparsely populated M94 system. Interestingly, the host galaxies in our simulations typically have higher stellar masses than the observed galaxies for which their LF best matches. This may be a result of observational incompleteness in measurements of stellar mass, which we could potentially reconcile by performing mock observations of our simulations and computing stellar mass in a comparable way.

Our simulations are not highly susceptible to the missing satellite problem, as we do not see systematic overproduction of luminous satellites. This is a result of the inclusion of baryonic physics in the simulation, which has been shown to reduce the number of satellites (Brooks et al. 2013). While each galaxy hosts on the order of 10-50 dark matter subhalos with $M_{\rm vir} > 10^8$ M$_\odot$, the occupation fraction for the least massive halos ($M_{\rm vir} < 10^8$ M$_\odot$) is low, with $< 5\%$ of halos hosting $> 10$ star particles.

Curiously, the most massive host, Sandra, hosts the greatest number of satellites in our simulations with baryonic physics, but in the corresponding DM-only simulation, its LF is significantly reduced in the massive ($M_{\rm vir} > 2 \times 10^9$ M$_\odot$) range. This disparity is particularly striking alongside the relatively low occupation fraction for Sandra. The inclusion of baryonic physics in this simulation appears to have increased the number of massive subhalos, but fewer of these host stars than is expected. Further analysis, with a focus on matching halos between DM-only and baryonic runs, is necessary to determine the precise cause of this discrepancy.

### 3.2. *Star Formation Properties*

We now investigate trends in the star formation properties of galaxies in both the Justice League and Marvelous simulations. Here, and throughout this paper, we define a "quenched" halo as one with a specific star formation rate SFR$/M_* < 10^{-11}$ yr$^{-1}$, where we define the SFR as the average rate of star-formation over the past 100 Myr. While many halos drop below this threshold only temporarily, most that do drop to sSFR $= 0$ yr$^{-1}$, indicating genuine periods of quiescence. With this definition, our sample shows an overall quenched fraction at $z = 0$ of $\approx 0.65$.

This choice of threshold has been adopted by many other studies using simulations (e.g. Wetzel et al. 2013; Bahé & McCarthy 2015; Pallero et al. 2019); however, observational studies often define quenching based on the atomic hydrogen richness $M_{\rm H\,I}/M_*$, a more easily measurable property. To facilitate observational comparisons, we compute the best-fit power-law relationship between the sSFR and atomic hydrogen richness in our simulations, finding that our sSFR threshold corresponds to $M_{\rm H\,I}/M_* \approx 0.2$.

Figure 3 shows one such comparison: the fraction of quenched satellites in each of the Justice League simulations, in bins of stellar mass, alongside observational data for the MW and M31 (from McConnachie 2012). For the observational data, we define the quenched fraction as the fraction of halos within 300 kpc of their host and with $M_{\rm H\,I}/M_* < 0.2$. Note that in the data of McConnachie (2012), this threshold produces identical quenched fractions as the threshold of 0.1 adopted by Fillingham et al. (2015) and Wetzel et al. (2015b), as the bimodal distribution of H I content means that the

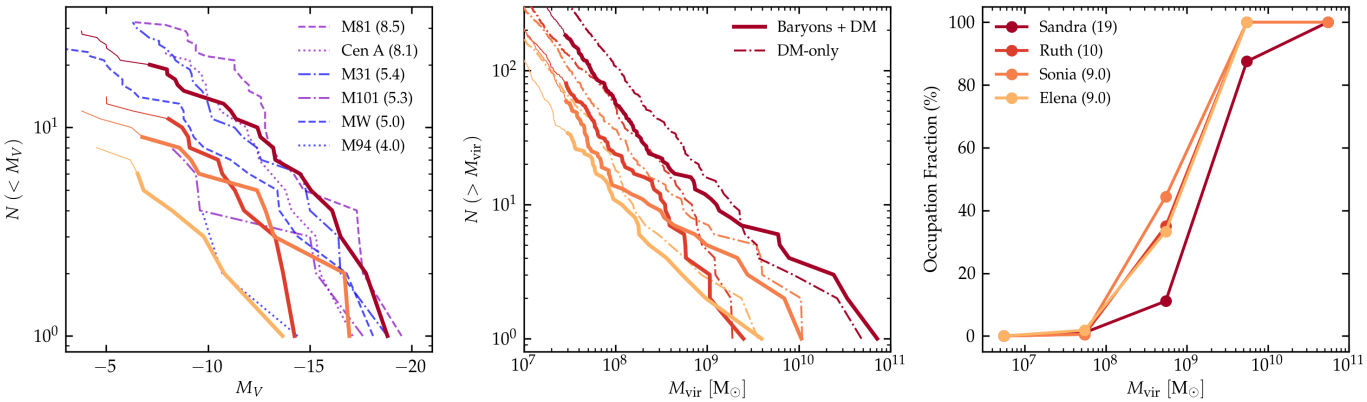


**Figure 2.** Left panel: Cumulative satellite luminosity functions for the Justice League simulations. Red/orange lines show simulated LFs, with thin lines extending below the high-resolution limit of 10 star particles. Observational luminosity functions, presented for comparison, are shown with blue/purple and in dashed lines. Observational data comes from McConnachie (2012) for the Milky Way and M31, Crnojević et al. (2019) for Centaurus A, and Bennet et al. (2019) for M101. For M94, we include the two satellites recently discovered by Smercina et al. (2018), along with two previously known satellites KK 160 and IC 3687. Data for M81 comes from Chiboucas et al. (2013) and includes the recently discovered satellite from Smercina et al. (2017), using the $M_{r'}$ to $M_V$ conversion provided by Crnojević et al. (2019). Values in parentheses represent the host galaxy's stellar mass, in $10^{10}$ M$_\odot$. Center panel: The corresponding cumulative satellite mass functions. In this case, the thin lines extend below the limit of 1000 total particles. The dot-dash lines are generated from equivalent dark matter-only simulations. Right panel: satellite occupation fraction, defined as the fraction of all halos in the baryonic simulations with more than 10 star particles, in bins of virial mass.

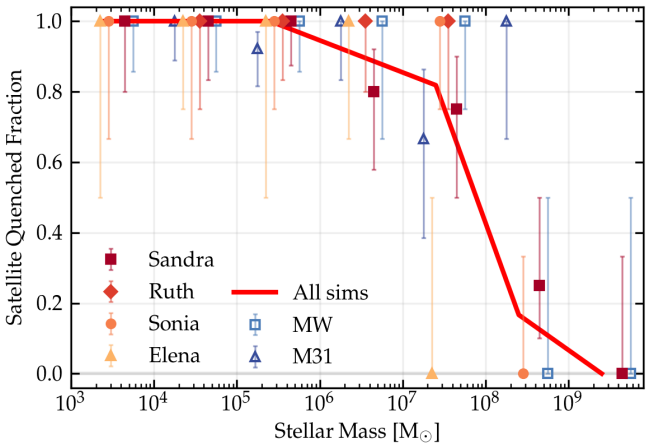


**Figure 3.** Satellite quenched fraction, in 1 dex bins of stellar mass. Individual simulations are indicated with points of varying shape, while the solid red line represents the quenched fractions across all four simulations. Observational data for the MW and M31 from McConnachie (2012) are shown in blue. Error bars represent 68% uncertainty in the binomial proportion via the Wilson Score Interval. We see considerable variation across simulations, particularly at $M_* = 10^{7-8}$ M$_\odot$, but this is consistent with similar variation between the MW and M31.

results are not sensitive to the exact choice of threshold. Additionally, the simulated quenched fractions in Figure 3 are qualitatively unchanged when using this H I-based definition, which simply lowers the fraction for Sandra at $M_* = 10^{6-7}$ M$_\odot$ from 0.8 to 0.6.

Figure 3 highlights key differences between the individual Justice League simulations. As with the satellite luminosity functions, we see considerable halo-to-halo scatter in the quenched fraction, particularly at stellar masses of $\sim 10^{7-8}$ M$_\odot$. This is consistent with variations across the MW and M31, where we see drastic differences in quenched fractions, particularly at higher stellar masses (i.e., the notably unquenched Magellenic clouds).

Figure 4 also shows the quenched fraction in bins of stellar mass, but instead divides the data between satellite and field dwarfs. Though the division between satellite and field is based on the distance to the nearest massive central galaxy (as discussed later), field galaxies are predominantly found in the Marvel-ous simulations. Observational data from Geha et al. (2012), McConnachie (2012), Karachentsev et al. (2013), and Weisz et al. (2015) are shown for comparison.

In the field, the quenched fractions in our simulations are highly consistent with observations. In particular, field dwarfs with $M_* > 10^7$ M$_\odot$ are entirely star-forming, as is observed by Geha et al. (2012). Below this stellar mass threshold, the quenched fraction rises steadily, suggesting that non-environmental quenching processes are efficient in this range. This increase at low masses closely follows morphology-based estimates of quenched fractions compiled by Weisz et al. (2015) using data from the Nearby Galaxy Catalog (Karachentsev et al. 2013). In compiling these data, Weisz et al.

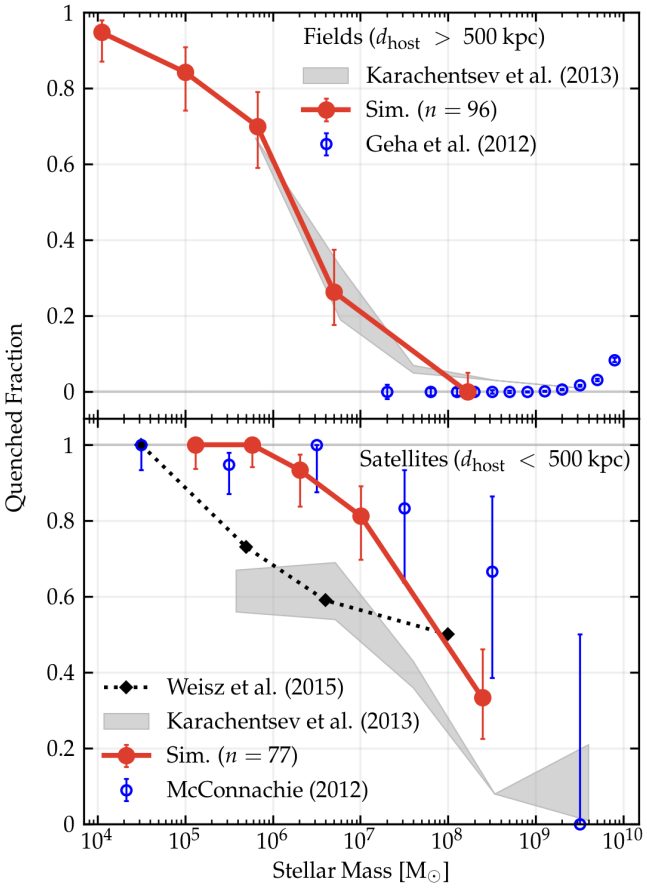


**Figure 4.** Fraction of quenched galaxies versus stellar mass. Top panel: field galaxies with $d_{\rm host} > 500$ kpc. Observational data from Geha et al. (2012) and Karachentsev et al. (2013) is presented for comparison. Bottom panel: satellite galaxies with $d_{\rm host} < 500$ kpc. Observational data from McConnachie (2012) (with error defined as in Figure 3), Weisz et al. (2015), and Karachentsev et al. (2013) is presented for comparison. In both panels, observational data from Karachentsev et al. (2013) is as compliled by Weisz et al. (2015). As in Figure 5, data is limited to galaxies with > 10 star particles and error bars represent 68% uncertainty in the binomial proportion via the Wilson Score Interval. Our data generally agree with observations of the Local Group and isolated dwarfs, and a clear dichotomy is apparent between satellites and field quenching at $M_* > 10^6$ M$_\odot$.

(2015) define a field galaxy as one > 500 kpc from the nearest host with $M_* > 10^{10}$ M$_\odot$; however, they define a quenched galaxy as one with morphological classification $T < 0$. The upper and lower limits on their quenched fractions indicate varying choice of whether to count dTrans galaxies as quenched. Given how different their definition of quenching is from ours, it is startling that our results agree so well, and suggest that morphological transition is well correlated with quenching.

For satellites, we generally see less agreement with quenched fraction observations (Figure 4, bottom panel). However, our simulations fit within the observed ranges, fitting the data from McConnachie (2012) well at low masses and Weisz et al. (2015) and Karachentsev et al. (2013) better at higher masses.

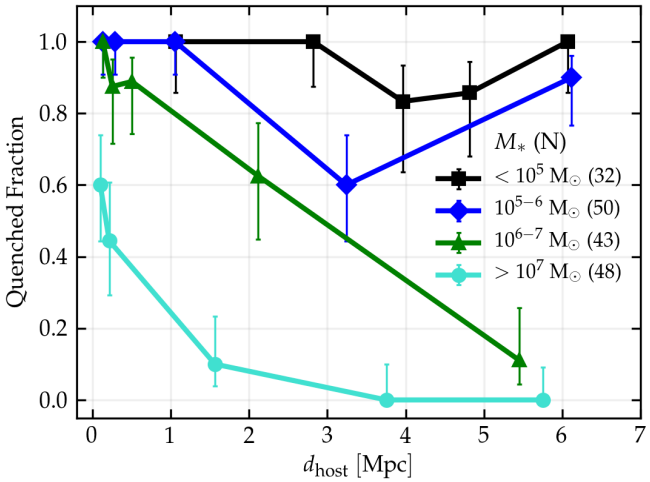


**Figure 5.** Fraction of quenched halos versus $d_{\rm host}$, the distance to the nearest massive host galaxy (i.e. with $M_{\rm vir} > 10^{11.5}$ M$_\odot$). The data is divided by stellar mass, with cyan circles representing "classical" dwarfs with $M_* > 10^7$ M$_\odot$, green triangles representing dwarfs in the range $10^{6-7}$ M$_\odot$, blue diamonds representing the dwarfs in the range $10^{5-6}$ M$_\odot$, and black squares representing the ultra-faint dwarfs with $M_* < 10^5$ M$_\odot$. Error bars represent 68% uncertainty in the binomial proportion via the Wilson Score Interval. This plot is restricted to only those halos with more than 10 star particles ($M_* \sim 10^4$ M$_\odot$). While quenching in massive ($M_* > 10^7$ M$_\odot$) dwarfs shows strong dependence on distance, this is not seen at the very lowest masses ($M_* < 10^6$ M$_\odot$). Interestingly, galaxies with $M_*$ in between these two show much larger-scale environmental trends, out to 4 Mpc from the nearest host.

Figure 5 shows the fraction of quenched galaxies versus distance to nearest massive host ($d_{\rm host}$), in bins of stellar mass. Here, and throughout this paper, we define a "massive host" as one with virial mass $M_{\rm vir} > 10^{11.5}$ M$_\odot$, roughly equivalent to the stellar mass threshold adopted by Weisz et al. (2015) and lower than the lowest-mass host halo in the Justice League. While a higher threshold of $M_{\rm vir} = 10^{12}$ M$_\odot$ would be more in line with the stellar mass threshold of $2.5 \times 10^{10}$ M$_\odot$ adopted by Geha et al. (2012), we have verified that our qualitative results do not change based on this choice of threshold. While the presence of a massive host is a given in the Justice League simulations, in the Marvelous simulations these massive hosts are DM-only halos that fall outside of the resolved region. Note that the $d_{\rm host}$ metric is used to distinguish between satellite and field dwarfs in Figure 4.

In the classical dwarf range ($10^7 < M_*/\mathrm{M}_\odot < 10^9$), we see quenched fractions begin to drop at distances beyond $\sim$ 300 kpc, the typical estimate for the virial radius of halos with $M_{\rm vir} \sim 10^{12}$ $\mathrm{M}_\odot$. In this mass range, the quenched fraction is identically 0 for dwarfs $\gtrsim$ 1 Mpc from the nearest massive host. This is in line with observations (Geha et al. 2012, their Figure 4), which show that nearly all classical dwarfs are star-forming beyond 1.5 Mpc.

However, at lower masses ($10^6 < M_*/\mathrm{M}_\odot < 10^7$), environmental quenching appears to extend far beyond 300 kpc, maintaining a quenched fraction of $\sim$ 0.6 even 2 Mpc from the nearest massive host and showing downward trends as far as 5 Mpc (Figure 5). This same effect can also be seen, albeit less strongly, in halos with $M_* < 10^6$ $\mathrm{M}_\odot$, where we remarkably find some star-forming dwarfs at high $d_{\rm host}$. These results are supported by recent observations of isolated, star-forming, dwarf galaxies at very low masses ($M_* < 10^6$ $\mathrm{M}_\odot$) where satellites are predominantly quenched (e.g. McQuinn et al. 2015). Of these faint, isolated ($d_{\rm host} > 1$ Mpc) dwarfs, $\sim$ 90% (65%) of those that are star-forming are found more than 100 kpc (300 kpc) from the nearest halo of any mass, indicating that the ability of these low-mass galaxies to form stars is additionally dependent upon the level of isolation from *any* other galaxies. In a hierarchical universe, matter will tend to cluster near the massive central galaxies (Moore et al. 1999), so $d_{\rm host}$ remains a good metric for assessing the level of isolation. These large-scale effects may be also representative of the role of group preprocessing at this mass scale, in which interactions with other low-mass galaxies may speed up the quenching process (e.g. Bahé et al. 2013; Wetzel et al. 2015a; Jung et al. 2018).

Additionally, Figure 5 shows universally high quenched fractions in the ultra-faint range ($M_* < 10^6$ $\mathrm{M}_\odot$) relative to higher-mass dwarfs. The majority ($\sim$ 80%) of fields and all satellites in this range are quenched, with the lowest quenched fractions at distances of $\sim$ 3–4 Mpc. The high quenched fractions and lack of significant correlation with distance indicates that the primary processes driving quenching in this range are not environmental. Particularly at $M_* < 10^5$ $\mathrm{M}_\odot$, reionization in the early universe has been shown to rapidly quench galaxies, leaving behind "true fossils" with no active star formation and universally ancient stellar populations (Ricotti & Gnedin 2005; Brown et al. 2014; Weisz et al. 2014). Reionization likely plays a role in quenching these ultra-faint dwarfs in our simulations, as $\sim$ 60% (90%) of our galaxies in this mass range have formed no stars in the past 11.6 (10) Gyr. While further discussion of quenching timescales is left for §4, it is clear that reionization may produce the high quenched fractions at this mass scale, either directly or by preprocessing galaxies for quenching via gas depletion or stellar feedback.

The dichotomy of quenching in satellite and field dwarfs is readily apparent in our simulations. While few quenched galaxies with $10^7$ $\mathrm{M}_\odot < M_* < 10^9$ $\mathrm{M}_\odot$ are found in the field, we see quenched fractions of $\sim$ 50–60% for satellites in this same mass range. This, dichotomy, alongside the variation across distance, is indicative of the role of the environment in suppression of star formation. However, it remains unclear the relative importance of various quenching processes, such as ram pressure stripping, tidal stirring, strangulation, feedback, and reionization. To better constrain the roles of these processes, we now investigate the quenching timescales of satellite galaxies in the Justice League simulations.

## 4. QUENCHING TIMESCALES

Having established the environmental dichotomy in quenching at higher masses, and the ubiquity of quenching at lower masses, we now endeavor to determine the processes driving this quenching. The relative times of quenching and infall can provide a useful method to assess the role of environmental quenching. Additionally, the quenching timescale, commonly defined as the time a satellite remains star-forming after infall, effectively measures the efficiency of environmental quenching and may hint at the underlying quenching processes.

### 4.1. *Timescale Definitions*

We define the quenching time ($t_{\rm quench}$) as the lookback time at which a galaxy had formed 99% of its present-day stellar mass. While many studies (e.g. Weisz et al. 2015; Simpson et al. 2018; Fillingham et al. 2019) use a threshold of 90%, this choice is intended to minimize observational uncertainty resulting from the presence of blue straggler stars (BSs). The ages of BSs are difficult to constrain using color-magnitude diagrams, and can contribute $\sim$ 5–10% to a galaxy's stellar mass (Weisz et al. 2015). In the absence of SFH uncertainty, we find that a threshold of 99% more accurately indicates the quenching epoch in our simulations. However, adopting the 90% threshold does not qualitatively change our results, simply shifting the quenching times earlier by a mean value of $1.3 \pm 1.5$ Gyr.

To determine $t_{\rm quench}$, we compute each halo's cumulative SFH from its $z = 0$ stellar content. We use the $z = 0$ stellar content intentionally, as opposed to the instantaneous SFR of the main progenitor, to facilitate more direct comparisons to observations (e.g. those in

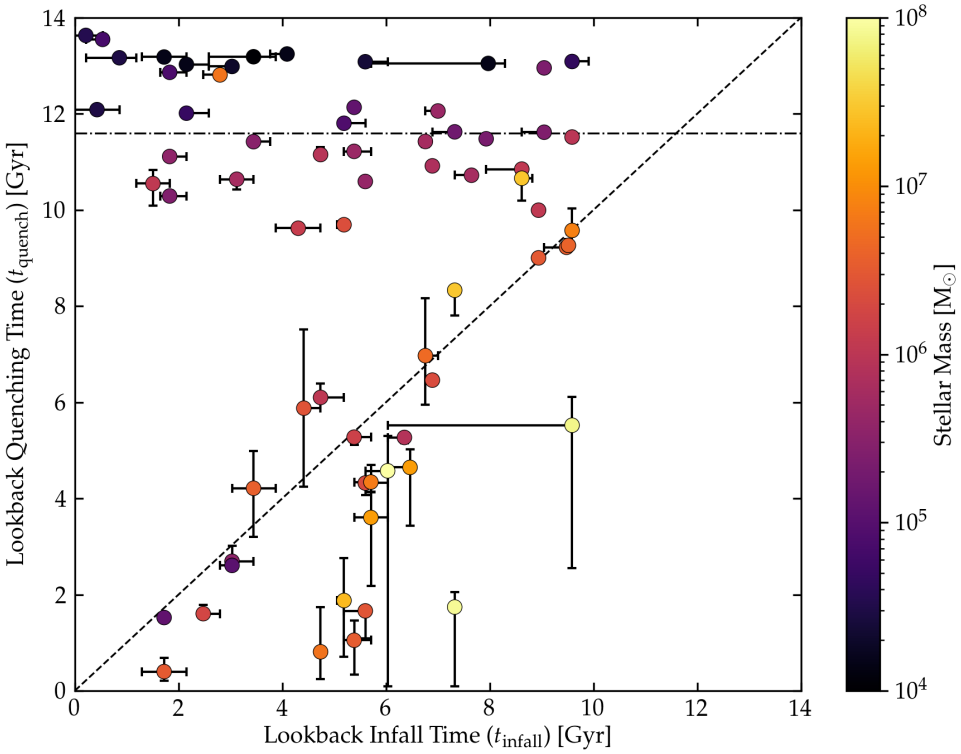


**Figure 6.** Lookback quenching time ($t_{\rm quench}$) vs lookback infall time ($t_{\rm infall}$) for all quenched halos surviving to $z = 0$ in our MW-analog simulations. Error bars indicate sensitivity to choice of threshold; that is, upper (lower) limits for quenching time use 90 (100)% rather than 99%, and upper (lower) limits for infall time use 1.1 (0.9) $R_{\rm vir}$ rather than 1 $R_{\rm vir}$. Points are colored by stellar mass, and the dot-dash line indicates the approximate time of the end of of the effects of reionization. Two major groups can be seen: those that quench early and non-environmentally (along the top) and those that quench at or after infall (along and to the right of the diagonal line).

Fillingham et al. 2019). Uncertainties in $t_{\rm quench}$ are computed from the difference between quenching times with different choices of threshold, namely 100% and 98%. This is intended to flag those galaxies with particularly "strung-out" star formation histories, for which the exact quenching epoch is difficult to constrain.

We define the infall time ($t_{\rm infall}$) as the lookback time to the subhalo's first crossing of its host's virial radius ($R_{\rm vir}$). While many halos fall within 1 $R_{\rm vir}$ multiple times, we chose the first crossing because the traditional picture of stripping-dominated quenching attributes the majority of gas removal to the first pericentric passage (Slater & Bell 2013). As with $t_{\rm quench}$, uncertainties are computed by varying the choice of threshold, namely from 0.9 to 1.1 $R_{\rm vir}$.

Finally, we use two metrics to assess the times at which gas is lost from the satellite. First, we define the gas-loss time ($t_{\rm gas}$) as the lookback time at which the gas fraction $M_{\rm gas}/M_{\rm baryons}$ falls below 10%. We choose this threshold because the gas fraction does not typically rise after crossing this point. Second, we define the H I gas-loss time ($t_{\rm H\,I}$) as the time at which the atomic hydrogen richness $M_{\rm H\,I}/M_*$ drops below 0.2 and remains suppressed to $z = 0$. Note that $M_{\rm H\,I}/M_* = 0.2$ is roughly equivalent to our threshold for quenching, as discussed in section 3.2.

### 4.2. *Environmental Quenching*

Figure 6 shows $t_{\rm quench}$ vs $t_{\rm infall}$ for all quenched halos surviving to $z = 0$ in the Justice League simulations. Galaxies falling to the right of the diagonal line on this plot are quenched after infall, while those to the left are quenched prior to infall. Importantly, we see a population of predominantly higher-mass halos that quench after infall into the host halo (along and to the right of the diagonal line). This mass-dependence is more readily apparent in Figure 7, which shows the quenching timescale $t_{\rm infall} - t_{\rm quench}$ vs stellar mass. Galaxies which have become satellites but remain star-forming at $z = 0$ are shown on this plot, as their "quenching" timescale can be interpreted as a lower limit on the potential timescale. For intermediate-mass satellites ($M_* \sim 10^{6-7}$ M$_\odot$), we predominantly see rapid quenching timescales ($\lesssim 2$ Gyr) consistent with observational estimates of LG timescales (Wetzel et al. 2015b; Filling-

ham et al. 2019). At higher masses ($M_* \gtrsim 10^8$ M$_\odot$), none of our satellites are quenched at $z = 0$, consistent with the unquenched Magellanic clouds, which experienced infall $\sim 1.5$ Gyr ago (Kallivayalil et al. 2013, their Figure 11). Additionally, our lower limits are largely below the quenching timescales estimated by Wheeler et al. (2014).

Figures 6 and 7 also show a group of satellites that quench shortly before infall into their host halo but well after the epoch of reionization (just to the left of the diagonal line and with $t_{\rm quench} \lesssim 8$ Gyr). Of the 5 satellites in this group, 1 is a reionization fossil that experienced momentary reignition prior to infall, and 3 have SFHs that extend beyond the time of infall, i.e. they continue to form stars slowly until being quenched environmentally (as reflected in their error bars in Figure 6). Had we used the lower $t_{\rm quench}$ threshold, these galaxies and more would definitively quench prior to infall, indicating that environmental processes may extend beyond 1 $R_{\rm vir}$. This extended environmental quenching is likely caused by some combination of unique orbital trajectory (e.g. infall along a gas filament), ram pressure stripping (even beyond the virial radius), and gas depletion (Bahé et al. 2013). Future work is needed to assess these unique cases, particularly in relation to outflows, stellar feedback, and gas-depletion timescales.

Finally, a clear stellar mass threshold of $10^8$ M$_\odot$ is apparent in Figure 7, above which satellites can remain star-forming for as much as 5 Gyr after infall. Below this threshold, the only galaxies that remain star-forming are those which have only recently crossed $R_{\rm vir}$, and may still quench within the expected timescale. This characteristic stellar mass is highly consistent with those found in other studies (e.g. Wheeler et al. 2014; Fillingham et al. 2015; Phillips et al. 2015), suggesting that while satellites with $M_* > 10^8$ M$_\odot$ are resistant to rapid environmental quenching, those with $M_* < 10^8$ M$_\odot$ are more sensitive to these processes. However, this is in conflict with the recent observations by the Satellites Around Galactic Analogs (SAGA) survey (Geha et al. 2017), which find that 26 of 27 satellites are actively star-forming, despite sampling numerous galaxies with $M_* < 10^8$ M$_\odot$. This discrepancy may indicate unique properties of the Local Group environment, and the complete SAGA survey results will provide valuable constraints on simulations of dwarf satellites.

### 4.3. *Reionization Quenching*

Figure 6 also shows many lower-mass halos that quench before infall, with nearly all galaxies with $M_* < 10^5$ M$_\odot$ quenched more than 11.6 Gyr ago, consistent with quenching due to reionization (Brown et al. 2014). The two sole exceptions to this are low-mass halos that quench significantly later than most, only $\sim 2$ Gyr ago (the two purple points in the lower left corner of Figure 6). One is a case of reignition—a reionization fossil that moved through a relatively dense filament of gas $\sim 6$ Gyr ago and restarted star formation, which continued until infall (much like in Wright et al. 2019). The other is similar, first forming stars $\sim 4$ Gyr ago due to sudden formation of H I gas. Further discussion of these unique SFHs is left for Appendix A.

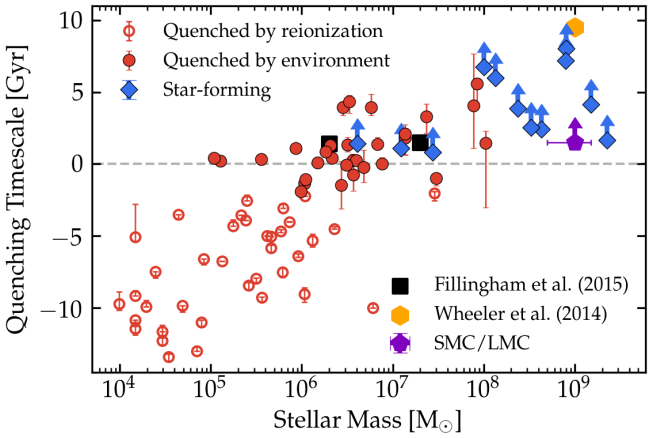


**Figure 7.** Quenching timescale ($t_{\rm infall} - t_{\rm quench}$) vs stellar mass. For halos that are quenched at $z = 0$ (red circles), this represents the time after infall until quenching, i.e. negative values indicate quenching prior to infall. For halos that are star-forming at $z = 0$ (blue diamonds), this can be interpreted physically as a lower limit on the possible quenching timescale. Halos quenched by reionization (i.e. $> 2$ Gyr before infall and $> 9$ Gyr ago) are shown in open points, while those quenched environmentally are shown in solid points. Error bars indicate threshold variation as in Figure 6, added in quadrature. Observational estimates from Fillingham et al. (2015) and Wheeler et al. (2014) are shown in black squares and orange hexagons, respectively. An estimate of the quenching timescale for the SMC/LMC system is shown in purple, using the infall time estimate from Kallivayalil et al. (2013). A clear stellar mass threshold of $\sim 10^8$ M$_\odot$ is apparent, above which our satellites are largely resistant to rapid environmental quenching.

Additionally, there is a large group of intermediate-mass halos which quench after reionization, but not due to infall into the host halo. These halos are likely able to survive reionization due to their higher halo masses, but unable to continue forming stars for long after. Another notable outlier is the sole reionization fossil with $M_* > 10^5$ M$_\odot$. This galaxy experiences no mergers in its lifetime, and does in fact form nearly $10^7$ M$_\odot$ of stars prior to the end of reionization. Further discussion of these cases is left for section 5.2.

## 5. DISCUSSION: QUENCHING MECHANISMS

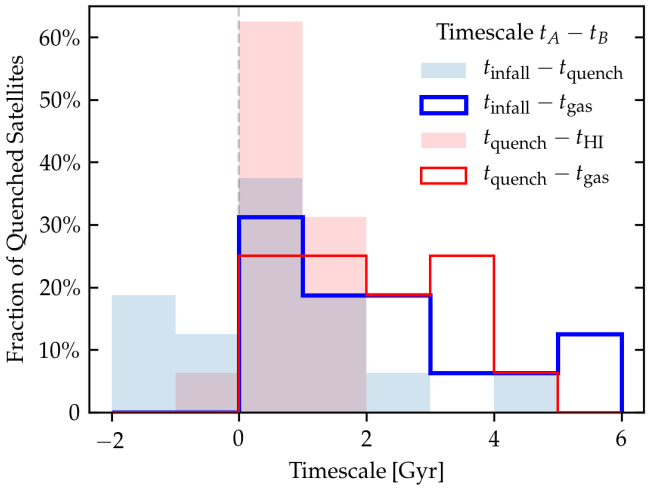


**Figure 8.** Histograms of timescale differences of the form $t_A - t_B$. Infall and quenching times are as defined previously, $t_{\rm gas}$ is the lookback time at which the satellite's gas fraction dropped below 10%, and $t_{\rm H\,I}$ is the lookback time at which the H I richness dropped below 0.2. Galaxies accounted for here are only that quench environmentally (i.e. not due to reionization) and have lost their gas by $z = 0$. Positive values indicate $t_A$ occurring before $t_B$. Here we can see clearly that while 70% of galaxies quench within 1 Gyr of H I gas-loss ($|t_{\rm quench} - t_{\rm H\,I}| < 1$), only 25% quench within 1 Gyr of gas loss ($|t_{\rm quench} - t_{\rm gas}| < 1$).

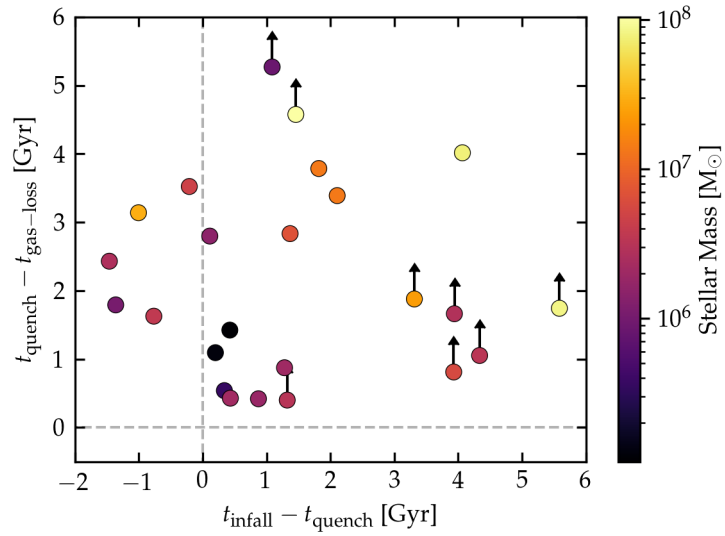


**Figure 9.** Gas-loss timescale $t_{\rm quench} - t_{\rm gas}$ vs quenching timescale $t_{\rm infall} - t_{\rm quench}$ for all satellites that quench environmentally. The quenching timescale represents the time after infall until quenching, and the gas-loss timescale represents the time after quenching until gas loss. Galaxies which retain their gas at $z = 0$ are shown as lower limits on this plot. While many satellites quench rapidly after infall (clustered in the lower left), there are many which quench rapidly but retain their gas for $\gtrsim 3$ Gyr (upper left) or quench $\gtrsim 3$ Gyr after infall and have not lost their gas at $z = 0$ (lower right).

### 5.1. *Rapid Environmental Quenching*

The rapid quenching timescales of galaxies in our sample (see Figure 7) implies the role of an environmental process that is highly efficient at quenching galaxies with $10^6 < M_*/{\rm M}_\odot < 10^8$. A likely candidate is ram pressure stripping, which has been posited to rapidly remove gas from infalling satellites (Slater & Bell 2014; Wheeler et al. 2014; Wetzel et al. 2015b). In order to investigate the role of ram pressure in environmental quenching, we look at the relative times of quenching, infall, and gas-loss. In this section, we focus on the subsample of galaxies which are quenched by $z = 0$ and are quenched environmentally (i.e. the filled red points in Figure 7).

Figure 8 shows density histograms of several important timescales for galaxies in this sample: the time a satellite remains star-forming after infall ($t_{\rm infall} - t_{\rm quench}$), the time a satellite maintains its gas after infall ($t_{\rm infall} - t_{\rm gas}$), the time a satellite maintains its gas after quenching ($t_{\rm quench} - t_{\rm gas}$), and the time a satellite maintains its H I gas after quenching ($t_{\rm H\,I} - t_{\rm gas}$). Recall that we define $t_{\rm gas}$ as the lookback time at which the satellite's gas fraction drops below 10%, and $t_{\rm H\,I}$ as the lookback time at which $M_{\rm H\,I}/M_*$ drops below 0.2 and remains suppressed to $z = 0$. In our sample of galaxies which quench environmentally, $\sim 50\%$ (90%) quench within 1 (2) Gyr of infall. However, only $\sim 30\%$ (50%) lose their gas within 1 (2) Gyr of infall. Similarly, only $\sim 25\%$ lose their gas within 1 Gyr of quenching, indicating that total gas-loss is unable to account for quenching in every satellite in this sample. However, H I gas-loss is highly correlated to quenching, with all satellites in the sample losing their H I within 2 Gyr of quenching. This dichotomy between H I and total gas-loss indicates an environmental quenching process that is able to remove star-forming gas without removing the entire gas reservoir in $\sim 50$–75% of the sample.

Figure 9 shows the gas-loss timescale $t_{\rm quench} - t_{\rm gas}$ vs the quenching timescale $t_{\rm infall} - t_{\rm quench}$ for all satellites that quench environmentally. There is a subsection of our sample ($\sim 30\%$) in which gas-loss, infall, and quenching all occur within 2 Gyr of each other, indicating rapid ram pressure stripping that removes all of the satellite's available gas. However, many of the more massive satellites in this sample quench rapidly after infall, but still retain their gas for $\gtrsim 3$ Gyr after quenching (i.e. $t_{\rm quench} - t_{\rm gas} > 3$). Other satellites are able to resist quenching for $\gtrsim 3$ Gyr after infall, and still maintain their gas by $z = 0$. This further indicates that an environmental process is at play that is able to quench satellites without removing their entire gas reservoir.

To determine the nature of this quenching, we investigate these environmentally quenched satellites on a case-by-case basis. Figure 10 shows representative time-series plots for two satellites in this sample—one in which quenching is driven by classical ram pressure stripping, and one in which quenching occurs before gas-loss. In both examples, H I gas-loss occurs at the time of quenching, which occurs at the first pericentric passage, but in the latter, the gas fraction remains nonzero until the second pericentric passage. Thus, this "slow" environmental quenching is due to the loss of the star-forming H I gas in the satellite's disk, likely the result of ram pressure combined with stellar feedback. The baryon cycle in these dwarf galaxies involves continuous supernovae-driven gas outflow, cooling, and reaccretion (Christensen et al. 2016; Anglés-Alcázar et al. 2017), and it has been well documented that ram pressure can interrupt this cycle by stripping the gas ejected by SNe (Caproni et al. 2015; Bahé & McCarthy 2015; Kazantzidis et al. 2017). These cases of slow ram pressure stripping occur primarily in higher mass halos ($M_* > 10^6$ M$_\odot$), but there are likely additional factors such as group preprocessing (Wetzel et al. 2015a) and orbital velocity at pericenter (Jung et al. 2018; Emerick et al. 2016) that determine whether ram pressure is capable of rapid or slow quenching.

Future work constraining the role of ram pressure stripping in our simulations will use the rates, directions, and temperatures of gas inflows and outflows to disentangle the roles of tidal stirring, ram pressure, and feedback in the quenching process (Engelhardt et al., *in prep*). Additionally, studies of satellite morphological transitions are necessary to assess the role of tidal effects in this interconnected gas-removal process.

### 5.2. *Quenching of Ultra-faint Dwarfs*

Quenching in the ultra-faint range has been the subject of many recent works, with particular focus on disentangling the role of environment and global processes (Rodriguez Wimberly et al. 2019; Fillingham et al. 2019). Reionization has been shown to efficiently quench the least massive dwarfs, whose low gravitational potentials are unable to resist the photoheating and ionization of gas. Observations show that 100% of star-formation in these "reionization fossils" is typically completed by 11.6 Gyr ago, with a stellar mass threshold of $\sim 10^5$ M$_\odot$ below which reionization dominates (Brown et al. 2014).

The near unity quenched fractions for ultra-faint dwarfs in our simulations (see Figures 4 and 5) indicated the role of a global quenching process, and Figure 6 shows a well-defined population of ultra-faint dwarfs which quenched very early in cosmic time. The

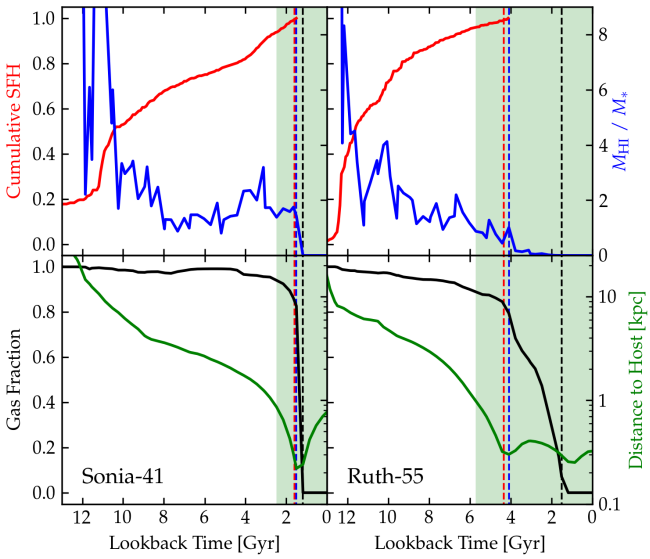


**Figure 10.** Time series plots for two representative subhalos: Sonia-41 on the left and Ruth-55 on the right. Top: cumulative star-formation history (red) and H I gas richness (blue). Bottom: distance from host (green) and gas fraction (black). The green shaded region shows the times at which the halo is within 1 virial radius of the host, and the red, blue, and black dotted lines indicate the times of quenching, H I gas-loss, and total gas-loss, respectively. The $z = 0$ stellar masses of Sonia-41 and Ruth-55 are $1.8 \times 10^6$ M$_\odot$ and $6.8 \times 10^6$ M$_\odot$, respectively. Sonia-41, on the left, is an example of rapid ram pressure-driven quenching, in which quenching occurs at the time of total gas removal (in this case at the first pericentric passage). Ruth-55, on the right, is an example of slow ram pressure, in which quenching occurs when H I gas is lost even though other gas still remains.

mean quenching time of all galaxies with $M_* < 10^5$ M$_\odot$ at $z = 0$ is $12.9 \pm 0.5$ Gyr ago, and as is readily apparent in Figure 7, these galaxies predominantly quench $\sim 10$ Gyr before infall. These may all be thought of as "true fossils" of reionization, as they form the vast majority of their stars prior to the end of reionization $\sim 12.5$ Gyr ago (Ricotti & Gnedin 2005). These "true fossils" make up 28% of the sample in Figures 6 and 7.

Several more massive halos quench shortly after the end of the documented effects of reionization, but far before infall into the host halo. To determine what distinguishes these from the true fossils of reionization, we plot the virial and gas masses of these halos at the end of reionization (12.5 Gyr) ago versus their quenching times (Figure 11). Galaxy mass at the end of reionization appears to be the predominant feature separating the survivors ($t_{\rm quench} < 12$) from those "true fossils" of reionization, with a critical threshold of $M_{\rm vir} \sim 4 \times 10^8$ M$_\odot$ or $M_{\rm gas} \sim 2 \times 10^7$ M$_\odot$. It is well documented that higher mass halos can more easily retain heated gas, al-

lowing $H_2$ self-shielding to maintain the high density, low temperature atomic gas in the core and extending star-formation until the H I gas is depleted (Glover & Clark 2012; Latif & Khochfar 2019; Chun et al. 2019). In lower mass halos, the hot, ionized hydrogen produced during this period is likely to escape the gravitational potential of the satellite and expose the dense, star-forming gas at the core to photoionizing radiation.

Figure 11 also helps to explain the outlying reionization fossil with $M_* \sim 10^7$ M$_\odot$, Ruth-33, whose mass is indicated with solid points. While its virial mass at the end of reionization is $\sim 2 \times 10^9$ M$_\odot$, well above that of others that quench at reionization, its gas mass is $\sim 4 \times 10^7$ M$_\odot$, inline with other fossils. It appears this galaxy's rapid, pre-reionization starburst left it with a depleted gas reservoir which was unable to self-shield and resist photoionization. As a sanity check, we highlight another notable outlier, Sonia-30, which survives reionization but has a stellar mass higher than most (see Figure 6). This galaxy is also indicated with solid points in Figure 11, and we see that its gas mass was particularly high at the end of reionization, unlike Ruth-33. Thus, it appears that high gas mass may be the more important factor in allowing dwarf galaxies to survive past reionization, and that $H_2$ self-shielding may be crucial to this process.

In both the survivors of reionization and in higher-mass galaxies which can resist ram pressure, gas depletion may play an important role in the quenching process. Overconsumption of star-forming gas combined with suppressed accretion has been shown to quench star formation, though on much longer timescales than ram pressure stripping (Wheeler et al. 2014). Future work is needed to assess the gas-depletion timescales of dwarf galaxies in these simulations.

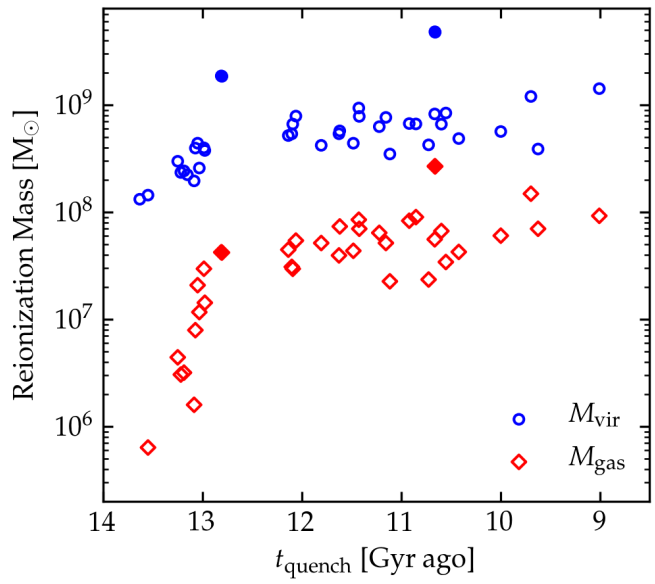


**Figure 11.** Mass at the end of reionization (12.5 Gyr ago) versus quenching time for the dwarfs that quench by reionization ($\gtrsim$ 9 Gyr ago). Blue circles show the virial mass, while red diamonds show the gas mass. The filled points represent notable outliers: Ruth-33 on the left and Sonia-30 on the right. We see a clear difference between the masses of the true fossils of reionization ($t_{\rm quench} > 12$) and the survivors.

## 6. CONCLUSIONS

Using the ultra-high resolution Justice League and Marvel-ous simulations, we have explored the quenched fractions and quenching timescales of dwarf galaxies near MW-mass hosts and in the field. In particular, our conclusions are summarized as follows:

1. Our simulations accurately reproduce observations of the Local Group, both in the satellite luminosity functions and quenched fractions. Notably, we find no quenched galaxies with $M_* > 10^7$ M$_\odot$ in the field, with a comparable fraction of $\sim$ 50% for satellites in this same mass range. At lower masses ($M_* < 10^6$ M$_\odot$), quenching is a ubiquitous phenomenon, occurring in nearly 80% of field and all satellite galaxies. While quenching varies considerably with distance to massive host in classical dwarfs, these ultra-faint dwarfs show less correlation. However, there appears to be much larger-scale environmental effects at play in intermediate-mass galaxies, potentially due to hierarchical structure and group preprocessing.

2. We find a stellar mass threshold of $\sim 10^8$ M$_\odot$, above which infalling satellites are largely resistant to rapid environmental quenching. Below this threshold, only three satellites remain star-forming, all of which only fell into the host halo $\lesssim$ 1 Gyr ago. While field galaxies below this threshold may remain star-forming, the majority of satellites are either quenched by reionization or quench rapidly ($\sim 1-2$) Gyr after infall.

3. We find a stellar mass threshold of $\sim 10^5$ M$_\odot$, below which galaxies are almost entirely quenched by reionization in the early universe. Above this threshold, there exists a straggler population which is not quenched directly by reionization, but likely depletes is supply of star-forming gas shortly thereafter. Notably, $H_2$ self-shielding appears crucial to allowing these galaxies to survive past the epoch of reionization, and we find that only galaxies with $M_{\rm gas} \gtrsim 2 \times 10^7$ at the end of reionization can survive.

4. We find that only $\sim 30\%$ of satellites that quench environmentally do so because of total gas-loss by ram pressure stripping. The remaining satellites quench on similar timescales, but maintain their gas for $\gtrsim 3$ Gyr after quenching. This is likely the result of ram pressure interrupting the satellite's baryon cycle and removing hot halo gas that would otherwise cool, reaccrete, and continue star-formation. Thus, it appears that stellar feedback plays an important role in environmental quenching of dwarf satellites.

There is much further work to be done in analyzing these simulations. Additional analysis of the observable properties (i.e. color, morphology) is in progress, for which preliminary results show good agreement with observations of the Local Group (Chamberland et al., *in prep*). Furthermore, the ultra-high resolution these simulations provide will facilitate detailed tracking of gas inflow and outflow, constraints on satellite contribution to the CGM, and analysis of observational signatures of quenching mechanisms. While it remains unclear the precise role that each hypothesized mechanism may play in the quenching of dwarf satellites, the Justice League simulations provide a promising route by which to better understand these processes.

We acknowledge Lucas Chamberland and Anna Engelhardt for fruitful discussions on the physics of dwarf satellite quenching and the numerous other summer research students in the Grinnell College Physics Department for their continuous support and enthusiasm. Additionally, we acknowledge the funding and support of the Grinnell College Mentored Advanced Project (MAP) program.

# APPENDIX

## A. CASE STUDIES IN REIGNITION

In their 2019 paper, Wright et al. explore dwarf galaxies with a unique type of "gappy" star formation history, characterized by long quiescent periods followed by the sudden reignition of SF. Many LG dwarfs have shown such star formation histories (namely Leo T, Leo A, ELM, and the Aquarius dIrr). By examining the histories of specific halos, Wright et al. (2019) conclude that reignition is caused primarily by relatively low-density, low-velocity interactions between the satellite and the CGM, in which ram pressure compresses (rather than strips) the hot gas in the halo. Many studies have explored the possibility of ram pressure and subsequent compression leading to rapid starburst (e.g. Fujita & Nagashima 1999; Henderson & Bekki 2016).

In our simulations, we see only one example of such a "gappy" star formation history: subhalo 31 in Elena. However, we also see an example of a similarly unique SFH: subhalo 36 in Sandra, a galaxy which first began forming stars $\sim 4$ yr ago. These two galaxies are notable outliers in Figure 6, as they quench far later than their stellar mass would suggest. In this section, we explore the possible causes of these outliers and compare to previous studies.

Figure 12 shows halo 31's cumulative SFH, along with various time-series gas properties including the atomic hydrogen fraction $M_{\mathrm{H\,I}}/M_*$, gas mass $M_{\mathrm{gas}}$, atomic hydrogen mass $M_{\mathrm{H\,I}}$, mean temperature, inflow rate, and outflow rate. We see a period (from $\sim 12$ Gyr ago to $\sim 6$ Gyr ago) in which the halo formed no stars, followed by a sudden reignition $\sim 6$ Gyr ago. Leading up to reignition, we see a rise in the H I gas fraction, peaking at the time of reignition and declining as gas is depleted. Interestingly, this rise is caused not by the accreation of new H I gas, but rather the conversion of existing gas into H I (as $M_{\mathrm{gas}}$ remains relatively constant during this period, there is no gas inflow, and the mean gas temperature drops as H I rises).

While this is consistent with ram pressure compressing the gaseous halo and leading to the formation of H I, we do not see a significant increase in the ratio of ram pressure to restoring pressure at this point. This is likely because we model the CGM as radially uniform in our ram pressure calculations, and thus do not take into account the possibility for movement through a filament of gas in the cosmic web. Density slices at various points in time corroborate this hypothesis, showing the halo passing through a dense region of gas and experiencing starburst. This is similar to subhalo 36 in Sandra, which did not have any stars prior to $\sim 4$ Gyr ago, as it did not have any H I gas. The lack of H I gas in this halo was likely due to its high temperature in its early life. As it began to cool, it passed through a filament of gas and accreted fresh H I, resulting in a starburst that continued until infall $\sim 2$ Gyr ago. A similar effect appears to be the cause of recent starburst in halo 36 in Sandra. These low-mass dwarfs with predominantly recent star-formation could make interesting targets for future observation.

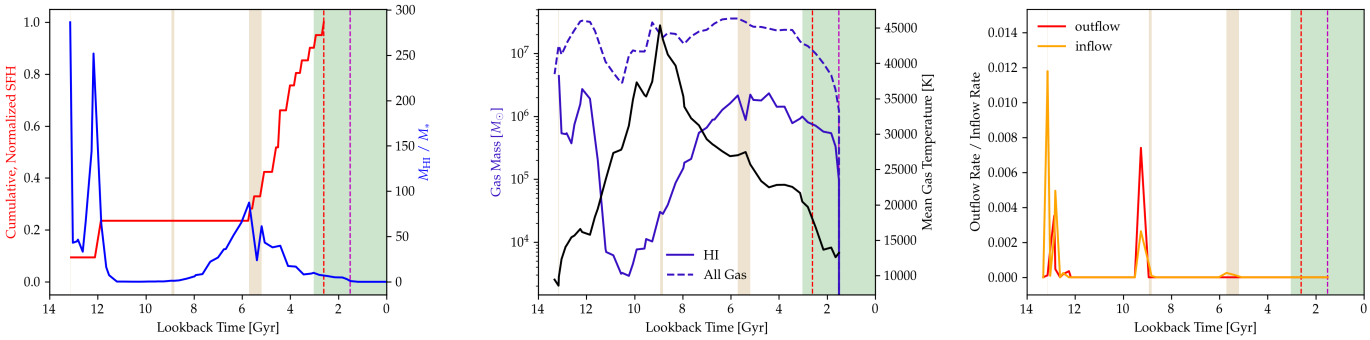


**Figure 12.** Time-series plots for subhalo 31 in Elena. Left panel: the cumulative star formation history and atomic hydrogen ratio $M_{\rm HI}/M_*$ over time. Center panel: the total gas and atomic hydrogen mass in the halo, and the mean gas temperature over time. Right panel: the gas inflow and outflow rates over time. Regions highlighted in green indicate times where the halo is within 1 $R_{\rm vir}$ of its host. Regions highlighted in tan indicate times in which the inflow rate exceeds the outflow rate, i.e. the halo is accreting gas. The vertical red line indicates the final SF shut-off point, and the vertical magenta line indicates a peak in the ratio of ram pressure to restoring pressure.

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